

Assumption of Linear Regression

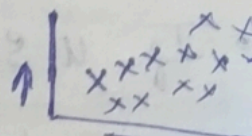
from Reg
analysis 1

Linear Regression relies on several assumptions to ensure the validity and reliability of the estimates and inferences. The key assumptions of linear regression are.

- i) Linearity
- ii) Normality of Residuals
- iii) Homoscedasticity

iv) No autocorrelation

v) No of little Multicollinearity



i) Linearity:-

The assumptions :- There is a linear relationship b/w the independent variable and the dependent variable.

→ The model assumes that changes in the independent variables lead to proportional changes in the dependent variable.

What happens when this assumption failed / violated :-

If the relationship is not linear, then

→ Biased Parameter Estimates

- The model will estimate wrong coefficients (slopes)
- This leads to wrong conclusions about how each feature affects the output.
- Ex:- You may think a feature has no effect, when it actually does.

→ Reduced Predictive Analysis Accuracy :-

- The model might under-fit, means fail to capture the true pattern.
- As a result, the predictions will be poor.
- Important trends & relationships in the dataset can be missed.

→ Invalid Hypothesis Tests and Confidence Intervals

- Statistical conclusions (like p-values, t-tests) become unreliable
- This affects the significance test & also affect sizes
- Decisions based on these results can be wrong.

How to check Linearity Assumptions :-

→ Scatter Plots :-

- Plot the dependent variable (y) against each independent variable ($x_1, x_2, x_3, \dots, x_n$)
- If the scatter plot looks like a straight-line pattern then linearity holds.
- If u see curves or patterns, linearity is likely violated.

→ Residual Plots :-

→ Residual = observed value (y) - Predicted value ($\hat{y}$)
 RSS (error)

→ Plot these residual against Predicted values or each independent variable

→ If linearity holds, residual should be

- randomly scattered around zero line
- without any systematic pattern

→ If you see curves/trends/clusters, linearity violated.

→ Polynomial Terms :-

Add Polynomial features (like x^2, x^3) and compare with the original model.

If the new model fits significantly better, it suggests the original (pure linear) assumption was wrong.

What to do if Linearity Fails:

Sometimes the relationship b/w X and Y becomes linear if we change their scale.

Goal: make the curve look more like a straight line.

i) Transformations:- apply to X values (features) or Y (target)

Common Transformations:-

i) $\text{Log}(\log X \text{ or } \log Y)$ $\rightarrow$ when data grows exponentially or varies widely.

ii) $\text{Square root}(\sqrt{x} \text{ or } \sqrt{y})$ $\rightarrow$ when the effect slows down as values increase

iii) $\text{Inverse}(1/x)$ $\rightarrow$ when high values of X reduces Y .

Ex:- If sales increase fast at low advertising spend but slow later, applying $\log(\text{Ad spend})$ may linearize it.

ii) Polynomial Regression:-

If the linearship is curved, not straight we can modify linear regression by adding powers of X .

Ex:- polynomial feature $\Rightarrow$
$$Y = b_0 + b_1X + b_2X^2 + b_3X^3$$

- It still uses linear regression, just with transformed features. & It can model curves like U, S-shaped pattern.
- Higher degrees (X^5, X^6) can overfit

iii) Piecewise Regression (Segment Regression)

The relationship changes at different ranges of X .

for Ex:- for $X < 50 \rightarrow$ one line pattern

for $X \geq 50 \rightarrow$ 2 line pattern

Why to use it:-

- Some data behaves differently at different levels.
- Instead of forcing one line through the entire data, we use multiple linear models in different segments

Real life Ex:- Relationship b/w income vs happiness

- from 0 to medium $\rightarrow$ happiness rises sharply
- after a point $\rightarrow$ happiness increases slowly.

Non-Parametric / Semi-Parametric Method

These methods do not force any specific shape (Linear or Polynomial). They let data decide the curve. (GAM), splines or kernel regression

When to use: -

When patterns are complex and randomly curved.

i) Normality of Residuals:

The assumption: The error terms (residuals) are assumed to follow normal distribution with a mean of zero and a constant variance

$$e_i \sim N(0, \sigma^2)$$

} reducible error

i) Inaccurate hypothesis Tests:

- Regression uses t-tests and F-test to check if coefficients are significant.
- These tests assume residuals are normally distributed
- If residuals (error) are not normal, these tests may give wrong P-values
- Result $\rightarrow$ you may wrongly conclude that a variable is significant or not significant.

ii) Invalid Confidence Intervals:

- Confidence interval for coefficients also assume normal residuals
- If residuals are not normal, the intervals become incorrect
- Ex:- A 95% CI may not actually represent 95% Confidence when normality is violated.

iii) Poor model Performance:

- Non-normal residuals often means the model is not capturing the pattern properly.

This may lead to:

- $\rightarrow$ reduced predictive accuracy
- $\rightarrow$ weaker model fit

The model might not be the right choice for the data.

iii) How to check the Normality of Assumption?

i) Histogram of Residuals:-

- Plot a histogram of residuals
- If it looks like a bell curve, residuals are normal
- If it looks skewed or has strange shapes → normality is violated.

ii) Q-Q Plot:-

- Compares residual quantiles with a normal distributed quantiles. If points lie along a straight line, residuals are normal.
- Deviations from the line → non-normality

iii) Statistical Tests:-

→ Omnibus test → Jarque-Bera test → Shapiro-Wilk

iv) What to do when assumptions fail:-

i) Apply Transformations

ii) Use Robust Regression methods:- Robust models reduce the effect of outliers and non-normal residuals

iii) Increase Sample size:- According to the Central Limit Theorem, as sample size increases → residual distribution becomes closer to normal.

→ with large datasets, normality is much less important.

iv) Switch to models that do not require Normality

Some models do not assume normality distributed errors.

Ex:-

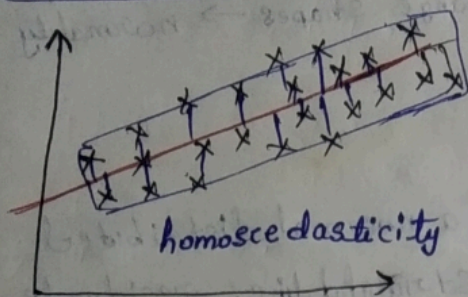
- Tree-based models → { Random Forest, XGBoost, Decision
- SVM Regression (SVR)
- K-NN Regression
- Neural Networks

v) Use Bootstrapping for Confidence Intervals and Hypothesis Tests.

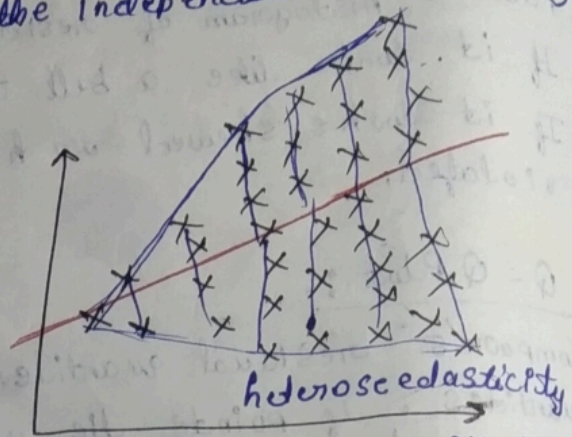
iii) Homoscedasticity:

The assumption: The variance (spread) of residuals is constant across all values of the independent variable (x)

residual = $y - \hat{y}$



→ Errors are consistent



→ Errors grow with x

when homoscedasticity is violated

i) Inefficient Estimates:

Even if heteroscedasticity happens, OLS still gives correct coefficients (unbiased). But

- Those coefficients are not efficient
- They are not the most accurate estimates
- They have higher variance than they should.

If the noise is constant → your measurement is stable

If the noise randomly increase or decreases → your measurement becomes shaky

→ That exactly what happens to β_0, β_1

ii) Inaccurate Hypothesis Tests: (conclusion becomes misleading)

→ T-tests & F-tests assume standard errors (SE) are correct & ~~var~~ variance of residuals is constant.

→ But under heteroscedasticity:

- Standard errors, p-value, t-test & f-tests becomes wrong

This causes:-

- a feature may look significant ($p < 0.05$) when it is actually not significant or it may look not significant when it is actually significant.

iii) Confidence Intervals :-

Confidence intervals depend on :

$$CI = \text{estimate} \pm (\text{critical value} \times \text{Standard Error})$$

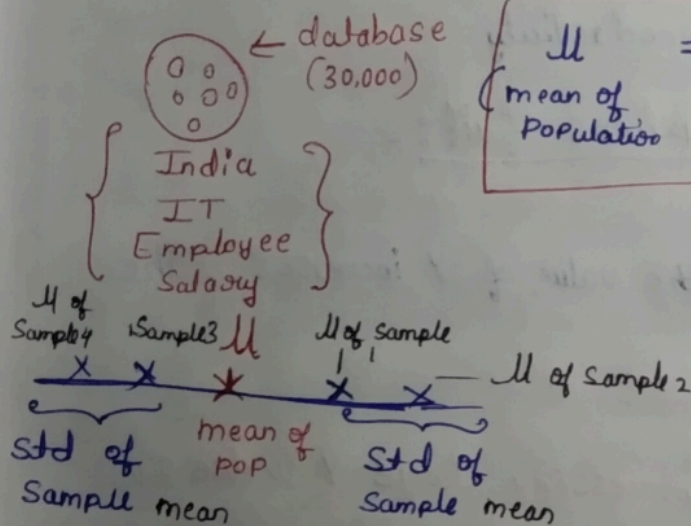
→ If SE is incorrect, the entire CI becomes wrong.

→ because of this, CI may look overconfident or underconfident.

* Standard Error :-

Standard Error (SE) measures how much an estimate (like a regression coefficient or sample mean) would fluctuate if we repeated the sampling process many times.

- standard deviation $\rightarrow$ variability in the data
- standard error $\Rightarrow$ variability in the estimate



$$\mu = \frac{\sum x_i}{n}$$

(mean of Population)

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

Standard deviation of the Sample mean is Standard Error

→ from Central Limit Theorem

$$N(\mu, \sigma/\sqrt{n})$$

$$SE = \frac{\sigma}{\sqrt{n}}$$

HQW Heteroscedasticity affects SE :-

Heteroscedasticity affects the Standard Error (S.E) because OLS assumes constant error Variance to compute S.E.

→ when residual variance changes across observations, the OLS formulas underestimate or overestimate the true variability of the coefficients.

→ This leads to incorrect standard error (SE), which produce wrong t-tests, p-values & CI.

iii) How to check the Homoscedasticity assumptions

i) Plot residual vs Fitted Values:-

X-axis $\rightarrow$ Predicted value ($\hat{y}$)

Y-axis $\rightarrow$ residuals ($e = y - \hat{y}$)

$\rightarrow$ Homoscedasticity:- Residuals are spread evenly

$\rightarrow$ Heteroscedasticity:- Residuals show pattern such as funnel shape, cone shape, u shape & increasing spread as $\hat{y}$ increases.

ii) Breusch - Pagan Test:- (Statistical test)

Tests if residual variance depends on fitted values

$\rightarrow$ p-value $< 0.05 \rightarrow$ heteroscedasticity

$\rightarrow$ p-value $> 0.05 \rightarrow$ homoscedasticity

iv) What to do when assumptions fail:-

i) Transform the target (y)

In many real datasets, as the value of y increases, the variance also increases.

Common Transformations:-

A transformation helps by compressing large y values, making variance more stable.

Log transformation $\Rightarrow \log(y) \rightarrow$ Compresses large values & shrink the distance b/w big y values

ii) Square root ($\sqrt{y}$)

iii) Box-Cox Transformation:

The coefficients themselves remain same. Homoscedasticity mainly affects the SE, not β -coefficients

ii) Robust Standard Errors (HCS / Huber-white SE)

$\rightarrow$ Normally OLS calculates (SE) assuming fixed variance

Robust SE recalculates (SE) in a way that does not assume constant variance. They only adjust $\Rightarrow$ SE,

t-value, p-values $\rightarrow$ CI

iv) Independence of Errors [No Auto correlation]

- Residuals (errors) should not be related to each other.
- Each error must be independent from the next error
- If errors are dependent: error shows a pattern like increasing, decreasing, cycles, auto pattern with time. Then independence fails
- This mostly happens in time series, stock prices

i) Why Independence is important :- If residuals are dependent

→ standard errors become wrong $\Rightarrow$ t-test, p-values become ~~un~~reliable.

→ F-test becomes invalid

→ model is missing important information

→ Confidence intervals get narrower

ii) How to detect :-

i) Residual vs Time Plot

if u see patterns $\rightarrow$ autocorrelation

ii) Durbin - Watson test :

• value near 2 $\rightarrow$ good

• value < 1.5 or $> 2.5 \rightarrow$ autocorrelation present

iii) ACF Plot (Autocorrelation Function) :-

Large spikes indicates correlation b/w residuals

iv) What to do if Independence fails

$\rightarrow$ use time series models

ARIMA, SARIMA, LSTM

- errors not random
- They follow sequence
- model is missing something

V) No Multicollinearity:-

Multicollinearity is a statistical phenomenon that occurs when two or more independent variables in a multiple regression model are highly correlated.

→ In other words, these variables exhibit a strong linear relationship, making it difficult to isolate the individual effects of each variable on the dependent variable.

When multicollinearity is bad

i) Inference = understanding relationships

Ex:-

- How much does X_1 affect Y ?
- Is X_2 statistically significant
- Which variable impacts Y the most
- What is the sign (+ve/-ve) of the relationship

Inferences uses:-

- Tests • P-values • Confidence Intervals
- Significance tests (F-test) • Interpretation of Coefficients

→ All of these become unreliable when multicollinearity exists.
Because when X_1 and X_2 are highly correlated

→ The model cannot separate their individual effects

→ Standard Error increases

→ Coefficients become unstable

$$Y = \beta_0 + \boxed{\beta_1} X_1 + \boxed{\beta_2} X_2$$

→ Signs may flip

→ P-values become wrong

• So for Inference tasks, multicollinearity is very harmful

Typical Inference Use Cases:-

- Economics
- healthcare analysis
- Academic research
- Marketing Analysis
- finance risk modelling

If X_1 & X_2 are correlated, changes done in X_2 will also impact X_1 , which leads to instability of coefficients.

ii) Prediction [multicollinearity is not bad]

- Prediction focuses on using a model to make accurate forecast or estimates for new unseen data.
- It aims to generalize the model to new instances, based on the patterns observed in the training data.

→ Predictions depend only on the final o/p $\hat{y}$, not on individual coefficients :- Even if multicollinearity makes coefficients unstable the model's predicted value $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ remains almost the same only the combined value matters for prediction.

→ When features are correlated, they carry the same information, so prediction accuracy does not drop :-

If x_1 & x_2 are highly correlated, the model still receives the same useful signal about y . So even if coefficients jump around, the info used for prediction remains constant.

→ Coefficients change but combination $\beta_1 x_1 + \beta_2 x_2$ stays stable.

• This is the key reason for prediction is unaffected.

if $x_2 \approx x_1$

model may estimate $\Rightarrow \beta_1 = 8, \beta_2 = -7$

Then prediction $\Rightarrow \hat{y} = 8x_1 - 7x_2$
 $\approx 8x - 7x = x$

→ The model compensates by redistributing weights among correlated features :-

If two features give similar information,

→ The model may reduce one coefficient and increase another

→ but their combined effect remains consistent.

• So the prediction stays accurate.

→ Prediction algorithm do not require Independent Features :-

→ ML models only need patterns to reduce error.

→ They do not care which feature gives the pattern.

as long as pattern → prediction accuracy stays good

Perfect Multicollinearity:

Perfect multicollinearity means one predictor is an exact linear combination of another.
 → There is an exact linear relationship b/w independent variables.

$$X_1 = a_1 X_2 + a_0 + \text{error}$$

$$X_1 = a_1 X_2 + a_0 \rightarrow \text{perfect multicollinearity}$$

Ex:

Cgpa	Percent	Ipa
9.12	91	7
8.5	85	6

$$\Rightarrow \text{Percent} = 10 \times \text{Cgpa} + 0$$

$$= 10 \times 8.5$$

$$= 85$$

$a_1 = 10$

$a_0 = 0$

Cgpa	Percent	Ipa
8	80	3
6	60	4

$$Ipa = \beta_0 + \beta_1 \text{cgpa} + \beta_2 \text{percent} + \text{error}$$

applying OLS method: $\beta = (X^T X)^{-1} X^T Y$

$$\begin{bmatrix} 1 & 8 & 80 \\ 1 & 6 & 60 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 \\ 8 & 6 \\ 80 & 60 \end{bmatrix} \begin{bmatrix} 1 & 8 & 80 \\ 1 & 6 & 60 \end{bmatrix}$$

$X \quad Y \quad X^T \quad X$

To compute Inverse calculate (determinant)

$$\text{Det} \Rightarrow 2(0) - 14(0) + 140(0)$$

$$\text{Det} = 0 \quad \left| \text{inverse} = \frac{1}{\text{det}} = \frac{1}{0} = \infty \right.$$

$$\begin{bmatrix} 2 & 14 & 1400 \\ 14 & 84 & 8400 \\ 140 & 8400 & 84000 \end{bmatrix}$$

→ Perfect multicollinearity makes $X^T X$ Singular (non-invertible)

→ A singular matrix has determinant = 0, so

inverse does not exist.

∴ OLS cannot compute coefficients

Multicollinearity Problems :-

- unstable coefficients (β)
- high standard error (X_1, X_2)
- difficult to determine which variable has the most impact on the dependent variable

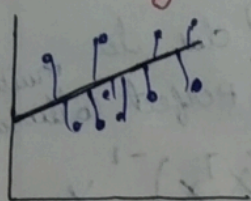
$$\beta = (X^T X)^{-1} X^T Y$$

$$\text{Var}(\beta) = \text{SE}(\beta)$$

for $\beta_0, \beta_1, \beta_2$

$$\text{SE}(\beta_0), \text{SE}(\beta_1), \text{SE}(\beta_2)$$

Simple Linear assumes that residual error follows normality.



⇒ Normal distribution

$$\Rightarrow N \sim (0, \sigma^2)$$

$$\text{Var}(\beta) = \sigma^2 (X^T X)^{-1}$$

Variance covariance matrix

$$\begin{bmatrix} a & - & - \\ - & b & - \\ - & - & c \end{bmatrix}$$

$$a \rightarrow \sqrt{\text{Var}(\beta_0)}$$

$$b \rightarrow \sqrt{\text{Var}(\beta_1)}$$

$$c \rightarrow \sqrt{\text{Var}(\beta_2)}$$

$$\text{SE}(\beta) = \sqrt{\text{diag}(\sigma^2 (X^T X)^{-1})}$$

(not possible)

i) when perfect multicollinearity

$$\text{determinant } (X^T X)^{-1} = 0$$

ii) When strong multicollinearity

$$\text{det}(X^T X) = \text{very very small or close to zero}$$

This leads to high standard error

Inflated Standard Error:-

$$\text{inverse} = \frac{1}{\text{det}}$$

$$\begin{bmatrix} a & - & - \\ - & b & - \\ - & - & c \end{bmatrix}$$

$$= \frac{1}{\text{very small num}} = \text{huge number}$$

unstable & unreliable Estimates:-

The regression coefficients become sensitive to small changes in the data, making it difficult to interpret the results accurately.

→ So that's why multicollinearity is bad for Inference
[since, estimation / coefficients are unstable, standard error is very high, we fail to explain each independent features on dependent (Y)]

Types of multicollinearity :-

i) Structural Multicollinearity :-

→ Multicollinearity happens because of how you created the features, not because of dataset.

Ex- adding polynomial features

→ Because of feature Engineering or model design introduces multicollinearity.

Ex- One hot Encoding

City

Delhi

Mumbai

Karnataka

OHE

D	M	K
1	0	0
0	1	0
0	0	1

} due to multicollinearity perfect

$$(X^T X)^{-1} X$$

$$\det = 0 \Rightarrow \text{no}$$

ii) Data-Driven Multicollinearity :-

→ Multicollinearity exists naturally in the data set, not because of feature Engineering.

→ The real-world data has variables that are naturally correlated.

How detect multicollinearity :-

Correlation
corr()

VIF

[Variance Inflation Factor]

Condition number

i) Correlation :- [Pairwise Correlation Check]

→ Checks Pairwise linear relationship b/w Predictors (X)

→ You compute a correlation matrix of all X variables correlation b/w two predictors

• |Correlation| > 0.8 → warning

Types of multicollinearity :-

i) Structural Multicollinearity :-

→ Multicollinearity happens because of how you created the features, not because of dataset

Ex- adding polynomial features

→ Because of feature Engineering or model design introduces multicollinearity.

Ex- One hot Encoding

City	D	M	K
Delhi	1	0	0
Mumbai	0	1	0
Karnataka	0	0	1

} due to perfect multicollinearity

$$(X^T X)^{-1} X$$

det = 0 $\Rightarrow$ no

ii) Data Driven Multicollinearity :-

→ Multicollinearity exists naturally in the data set, not because of feature Engineering.

→ The real-world data has variables that are naturally correlated.

How detect multicollinearity :-

Correlation
corr()

VIF
[Variance Inflation Factor]

Condition number

i) Correlation :- [Pairwise Correlation Check]

→ Checks Pairwise linear relationship b/w Predictors (X)

→ You compute a correlation matrix of all X variables
correlation b/w two predictors

• |correlation| > 0.8 $\rightarrow$ warning

• $| \text{correlation} | > 0.9 \rightarrow \text{Strong multicollinearity}$

$\rightarrow$ If ^{two} Independent variables show very high correlation, $\left\{ \begin{array}{l} \cdot \text{ They carry similar information} \\ \cdot \text{ will cause unstable coefficients in linear regression} \end{array} \right.$

ii) Variation Inflation Factor :-

VIF tells how much the ~~variation~~ variance of coefficients is inflated (increased) the predictor (x_i) correlated with other predictors

If multicollinearity increases $\rightarrow$ (SE) increases $\rightarrow$ Coefficients become unstable

Ex:-

	TV	NP	Radio	Sales
1st	-	-	-	
	x_1	x_2	y	
	x_1	y	x_2	

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

like this calculate ^{LR} for diff features & estimate R^2_{score}

How VIF works :- for Each Predictor x_i

Step 1:- Calculate Regression on Remaining feature, Except. Similarly do it for all the features.

Step 2:- Calculate R_i^2 -score of this regression

Step 3:-

$$VIF = \frac{1}{1 - R_i^2 \text{ score}}$$

$$\text{If } R^2 = 0.9$$

• If R_i^2 -score is high, x_i can be predicted by other predictors $\rightarrow$ bad multicollinearity

$$VIF = \frac{1}{1 - 0.9} = \frac{1}{0.1} = 10 \text{ bad}$$

• Then $1 - R_i^2$ becomes small

• Dividing by a small number $\rightarrow$ VIF becomes huge

High correlation among Predictors $\Rightarrow$ High VIF $\Rightarrow$ High standard Error

VIF

$1 \Rightarrow$ no multicollinearity

$1-5 \Rightarrow$ acceptable

$> 5 \Rightarrow$ concerning

$> 10 \Rightarrow$ serious multicollinearity